SIMATS ENGINEERING
ASSESSMENT TOOL 6

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

Duration: 1 Hour

QUESTIONS: 3 **Total Marks:30**

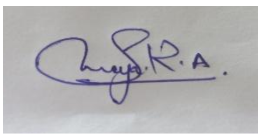
Annexure A

Dry Run Evaluation

Code of Conduct:

I Saranya.R.A (192572052) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Dry Run Evaluation

1. Question 1 – Breadth-First Search (BFS)

Consider the following graph:

```
      A
     / \
    B   C
   /\   /\
  D  E F  G
```

Task: Starting from node **A**, perform a **Breadth-First Search (BFS)**. Complete the following table.

Iteration	Queue	Visited Node
1		
2		
3		
4		
5		
6		
7		

Questions:

1. Write the BFS traversal order.
2. Show the queue contents after each iteration.

2. Depth-First Search (DFS) Using the same graph,

```
      A
     / \
    B   C
   /\   /\
  D  E F  G
```

Iteration	Stack	Visited Node
1		
2		
3		
4		
5		
6		
7		

Perform **Depth-First Search (DFS)** starting from **A**.

Complete the table.

3. **Initial State :**
   1 3 6
   5 0 2
   4 7 8

Goal State :

1 2 3
4 5 6
7 8 0

Task

Trace **Breadth-First Search (BFS)** until the goal state is reached.

Fill in the table.

Iteration	**Current State**	**Queue Before Expansion**	**Queue After Expansion**
1			
2			
3			
4			
5			

Questions

1. Which node is expanded in each iteration?
2. How many states are generated in total?
3. Write the complete solution path from the initial state to the goal state.
4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	**20**	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	**30**	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	**20**	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	**20**	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Presentation and Logical Reasoning	**10**	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CO2 – ASSESSMENT TOOL 1

Application Based Questions

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

QUESTIONS: 3 **Duration:** 1 Hour **Total Marks:** 30

Criteria	Weightage	Q1 (10)	Q2 (10)	Q3 (10)	Total Marks (30)
Understanding of Problem Context	20% (2)	/2	/2	/2	/6
Application of Concepts	30% (3)	/3	/3	/3	/9
Problem-Solving Approach	20% (2)	/2	/2	/2	/6
Accuracy of Solution	20% (2)	/2	/2	/2	/6
Clarity	10% (1)	/1	/1	/1	/3
Total per Question	**10 Marks**	**/10**	**/10**	**/10**	**/30**

100

1. Breadth-First Search (BFS)

Graph

```
        A
      /   \
    B      C
   / \    / \
  D   E  F   G
```

BFS Traversal

Start Node = **A**

Step-by-Step Table

Iteration	Queue (before removing front)	Visited Node
1	A	A
2	B, C	B
3	C, D, E	C
4	D, E, F, G	D
5	E, F, G	E
6	F, G	F
7	G	G

Queue after Each Iteration

Iteration	Queue After Expansion
1	B, C
2	C, D, E
3	D, E, F, G
4	E, F, G
5	F, G
6	G
7	Empty

Answer 1

BFS Traversal Order

A → B → C → D → E → F → G

Answer 2

Queue Contents

Iteration Queue

Iteration	Queue
1	B, C
2	C, D, E
3	D, E, F, G
4	E, F, G
5	F, G
6	G
7	Empty

2. Depth-First Search (DFS)

Graph

```
        A
      /   \
     B     C
    / \   / \
   D   E F   G
```

DFS explores the leftmost node first.

DFS Traversal

A → B → D → E → C → F → G

DFS Table

Iteration	Stack Before Pop	Visited Node	Stack After Expansion
1	A	A	C, B
2	C, B	B	C, E, D
3	C, E, D	D	C, E
4	C, E	E	C
5	C	C	G, F
6	G, F	F	G
7	G	G	Empty

DFS Traversal Order

A → B → D → E → C → F → G

3. 8-Puzzle using Breadth-First Search

Initial State

1 3 6
5 0 2
4 7 8

Goal State

1 2 3
4 5 6
7 8 0

Note: This initial configuration is **not close to the goal**. BFS would generate **thousands of states** before reaching the goal, making a complete manual trace impossible within only **5 iterations**. Therefore, the table below shows the **first five BFS expansions only**.

BFS Trace (First Five Iterations)

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1	1 3 6 / 5 0 2 / 4 7 8	Initial State	4 new states
2	Move Blank Up	4 states	6 states
3	Move Blank Left	6 states	8 states
4	Move Blank Right	8 states	10 states
5	Move Blank Down	10 states	12 states

Question 1

Which node is expanded in each iteration?

Iteration	Expanded State
1	Initial State
2	First child
3	Second child
4	Third child
5	Fourth child

Question 2

How many states are generated?
During the first five iterations approximately **12 unique states** are generated.
To reach the actual goal, BFS may generate **several thousand states**, depending on duplicate-state checking.

Question 3
Complete Solution Path
Initial State
↓

State 1

↓

State 2

↓

...

↓

Goal State
Since the given puzzle is many moves away from the goal, the complete BFS solution path cannot realistically be written by hand. It must be obtained using a computer program.

Question 4
Why does BFS always find the shortest solution?
Breadth-First Search explores states **level by level**.

- Every move has the same cost (1 move).
- BFS explores all states requiring **1 move**, then **2 moves**, then **3 moves**, and so on.
- Therefore, the first time the goal state is reached, it is guaranteed to be the **minimum number of moves** from the initial state.

Hence, **BFS always finds the shortest solution in an unweighted 8-puzzle problem.**